

SHE GUARD: A WOMEN CENTRIC HUMAN VIOLENCE DETECTION SYSTEM

HIBA P A (JEC22AD034)

HISHAM ALI (JEC22AD035)

MUHAMMED IRFAN (JEC22AD042)

RAZAAN RASHEED (JEC22AD049)

Supervised by: Mr SHINE P XAVIER



Department of Artificial Intelligence and Data Science
Jyothi Engineering College, Cheruthuruthy.

Vision of the Department

- Creating ethical leaders in the domain of Artificial intelligence and data Science through effectual teaching and learning process to develop emerging technology solutions for the benefits of industry and society with a focus on holistic learning and excellence

Mission of the Department

- Strengthening basic competencies in the domains of Artificial Intelligence and Data Science
- Providing high-quality, value-based technical education and developing technology professionals with creative ideas and compelling leadership abilities.
- Using logical thinking to create and develop cutting-edge products in collaboration with industry stakeholders in order to meet global expectations and requirements
- Enabling graduates to adapt to new technologies via strong fundamentals and lifetime learning

Programme Educational Objectives (PEOs)

- Graduates will demonstrate comprehensive expertise in artificial intelligence, machine learning, and data analysis while applying interdisciplinary knowledge in computer science, statistics, and advanced mathematics to address real-world challenges.
- Graduates will innovate and create impactful solutions ethically, contributing to societal betterment, leveraging their leadership, collaboration, and communication skills for effective teamwork and multidisciplinary endeavors.
- Continuously fostering a culture of lifelong learning, graduates will adapt to evolving technological landscapes, demonstrating adaptability and a commitment to ongoing professional development in the dynamic field of AI and data science.

Programme Outcomes

Engineering Graduates will be able to:

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

- **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

- **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

- **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

On the completion of Artificial Intelligence and Data Science program, the students will possess:

- Graduates will be able to apply core principles of science, engineering, and mathematics to develop AI-driven solutions for real-world challenges.
- Graduates will design, develop and implement efficient as well as scalable algorithms using programming languages, data structures and cloud computing technologies.
- Graduates will integrate ethical, social, and security considerations while developing AI-driven intelligent systems, ensuring responsible AI deployment.

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- Ensuring public safety is a growing concern, with traditional surveillance often failing to detect violence in real time.
- Our **Human Violence Detection System** uses AI and computer vision to identify violent behavior, such as fights and assaults, from video footage.
- By providing instant alerts, it enhances security, enables faster response times, and helps prevent crime in public and private spaces.

Problem Statement

Develop an intelligent system using deep learning models to detect violence in real-time from video feeds. Implement an algorithm for action recognition and tracking, and create an automated alert system for notifying authorities. Design the system to be scalable and integrate easily with existing surveillance infrastructure.

Objectives

- To develop an automated human violence detection system using YOLOv8s and OpenCV that can proactively identify violent activities in real-time surveillance footage.
- To reduce the dependency on manual monitoring in surveillance systems, making violence detection faster, more accurate, and more efficient in public and private spaces.
- To safeguard human safety and security by accurately detecting violent actions, thereby preventing potential crimes, injuries, and property damage.
- To utilize advanced deep learning techniques, specifically YOLOv8s, to achieve high accuracy in detecting violence in video streams and ensuring reliable and timely alerts through sound alarms and email notifications.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
One	Journal,IEEE	Integrated Violence and Weapon Detection Using Deep Learning,2024,Vaibhav Yadav,Sanskar Kumar,Atul Goyal,Sidharth Bhatla,Geeta Sikka,Amandeep Kaur [?]	A comprehensive detection system is prososed to identify violence and weapons.It leverages CNNs, LSTMs, and Vision Transformers to analyze visual data, capture relationships, and enhance feature representation.Utilizing TensorFlow or PyTorch, the system boasts high accuracy 98 percentage, real-time processing (131 fps), and versatility. .

Table 1: Literature Review

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Two	Journal,European journal of science and technology	Violence Detection with Machine Learning: A Sociodemographic Approach ,2023,Tolga Ensari,Betül Ensari,Mustafa Dagtekin [?]	A machine learning framework analyzes sociodemographic data to identify high-risk factors for domestic violence. The Decision Tree (DT) algorithm showed the best performance, achieving an accuracy of 55 percentage, enabling proactive prevention and resource allocation.However, limitations include reliance on dataset quality, potential lack of generalizability, and the model's accuracy being limited to 55 percentage.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Three	Journal,Journal of Physics: Conference SeriesE	Human Violence Detection Using Deep Learning Techniques,2022,S. A. Arun Akash,R. Sri Skandha Moorthy,K.Esha, N.Nathiya[?]	A deep learning framework utilizing CNNs (Inception-v3 and Yolo-v5) detects and classifies objects in video analysis. Simulation experiments on CCTV footage achieved 74 percentage accuracy in detecting violent acts. While enabling real-time monitoring for public safety, limitations include false positives/negatives and accuracy issues due to limited training data .

Table 3: Literature Review

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Four	Journal,IEEE	Design of Violence Event Detection System Based on CCTVs by Human Body Pose Recognition,2023, Kuei-Chung Chang,Yi-Ching Liao [?]	A human body pose recognition system detects violent behaviors through abnormal joint angles and movement speeds. Using OpenPose, it analyzes CCTV images for real-time skeleton recognition. Experimental results show 85 percentage accurate detection and automated alerts. it include real-time detection, non-intrusive monitoring, and timely intervention. Limitations are reliance on camera quality, risk of false positives

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Five	Journal,IEEE	Weapon Detection in Real-Time CCTV Videos Using Deep Learning,2021, Muhammad Tahir Bhatti, Muhammad Gufran Khan, Masood Aslam, Muhammad Junaid Fiaz [?]	A deep learning-based system for real-time weapon detection in CCTV footage is proposed, utilizing algorithms like VGG16, Inception-V3, YOLOv3, and YOLOv4. YOLOv4 achieved the best results, with an F1-score of 91 percentage and mean average precision of 90.73 percentage. The system offers high accuracy, real-time processing, and a custom dataset, but faces limitations including dataset constraints, occlusions.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Six	Journal,International Journal on Artificial Intelligence Tools	Fast Violence Recognition in Video Surveillance by Integrating Object Detection and Conv-LSTM,2022, N. Jain, Vandana Gupta, U. Tariq, D. Jude Hemanth [?]	The paper proposes a hybrid model, HWVd, for detecting violence in video surveillance. It integrates human detection, weapon detection, and spatial feature extraction using CNNs and LSTMs. The model achieves 91 percentage accuracy, offers real-time processing, and economical storage. However, potential limitations include performance in diverse environments and robustness against false positives.

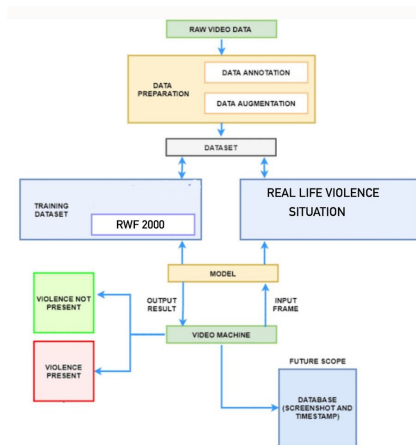
Gaps identified

- The accuracy and generalizability of violence detection models are limited due to the availability of less diverse and small-sized datasets, which restricts their ability to detect violence in complex real-world scenarios effectively.
- Some existing models exhibit low accuracy in real-time violence detection, which compromises their reliability and effectiveness in dynamic surveillance environments.
- Many current models are only capable of detecting limited types of violence activities, making them less adaptable to various forms of aggressive behavior.
- Most of the existing systems lack an automated alert mechanism (such as sound alarms or email notifications), which reduces their practicality for real-time security applications.

Proposed System

- **Dataset Collection:** Uses diverse video footage of violent and non-violent activities for training (e.g., RWF-2000)
- **Data Preprocessing:** Extracts frames, resizes (640x640), normalizes, and augments data for better model performance.
- **YOLOv8s Model Training:** Annotates violent actions, trains YOLOv8s with labeled data, and fine-tunes for accuracy.
- **Real-Time Detection Alerts:** returns violence if it is true and sends a sound alarm and alert email to user.
- **User Interface Design:** Develops a dashboard for real-time monitoring and alert management.

Architecture of Proposed System



Hardware Requirements

- NVIDIA GPU (GTX 1660 or higher) and Intel i5/i7 or AMD Ryzen 5/7 processor
- 16GB RAM
- 200GB storage
- Surveillance cameras or webcams

Software Requirements

- VS Code
- Python programming language for AI/ML development
- Ultralytics, PyTorch, OpenCV, NumPy, pandas

Dataset

Dataset :RWF-2000(from KAGGLE)

Source : The RWF-2000 dataset, sourced from Kaggle, contains 2,000 surveillance videos for violence detection, with 1,000 fight and 1,000 non-fight clips. It is split into 1,500 training videos (750 fight, 750 non-fight) and 500 testing videos (250 fight, 250 non-fight). Videos are in .avi format, 3-10 seconds long, with a frame rate of 25-30 FPS.



(a) Non-physical violence scene in AIRTLab



(b) Non-physical violence scene in Pexels



(c) Physical violence scene in RVLS



(d) Physical violence scene in SCVD

- **test accuracy:** SHE GUARD - Real-Time Human Violence Detection Model was trained using 25 epochs and a batch size of 8. The model achieved an accuracy of 80 percentage on the test set, indicating its proficiency in accurately detecting and classifying violent activities in real-time surveillance scenarios.

Result

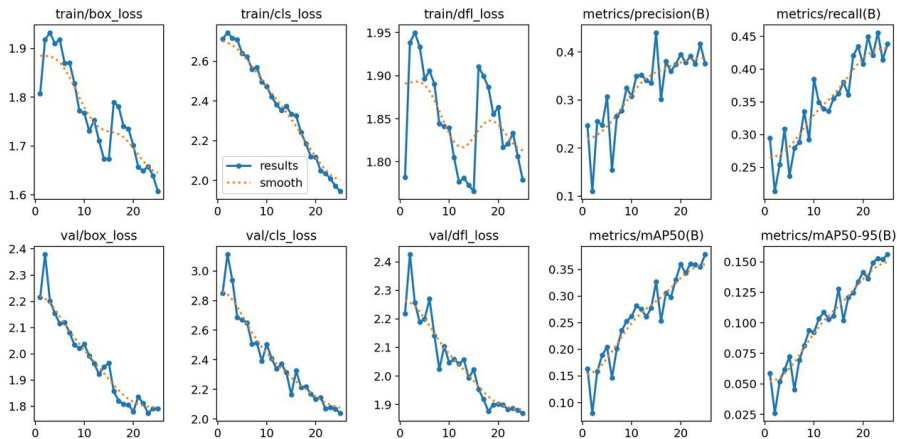


Figure 4: graphs for accuracy and loss

- from fig 4 the graph shows the training and validation performance of SHE GUARD, a real-time human violence detection system.
- Training losses (box, class, DFL) consistently decrease, indicating effective model learning.
- Validation losses also decrease, showing good generalization and no overfitting. Precision increases over time, reducing false positives.
- Recall improves steadily, capturing more violent incidents. mAP50 and mAP50-95 scores rise, reflecting better object detection accuracy. Overall, the model demonstrates strong learning and reliable real-time violence detection performance.
- **Real-Time Performance Evaluation:** The SHE GUARD system was tested at different frame rates — 3 FPS, 5 FPS, and 10 FPS. It was observed that increasing FPS reduced detection accuracy due to processing limitations and frame overlap issues. The optimal performance was achieved at 3 FPS, balancing detection accuracy and real-time efficiency.

User Interface

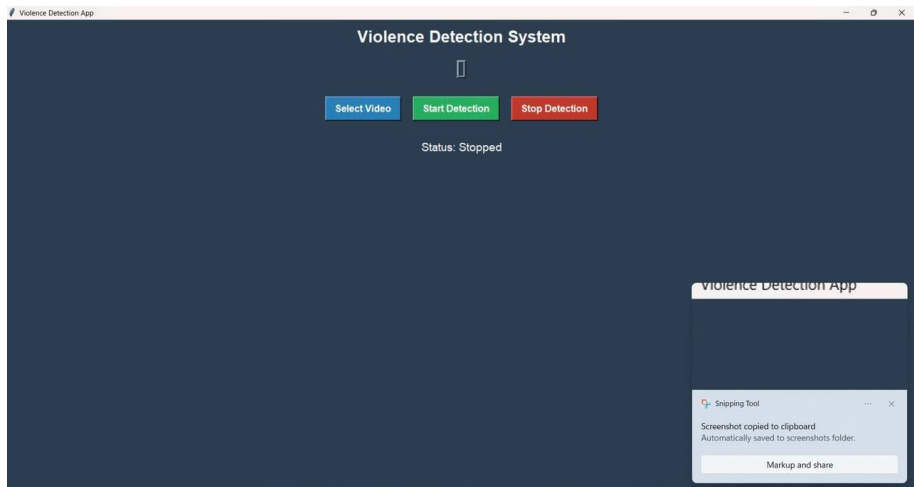


Figure 5: opening interface

User Interface

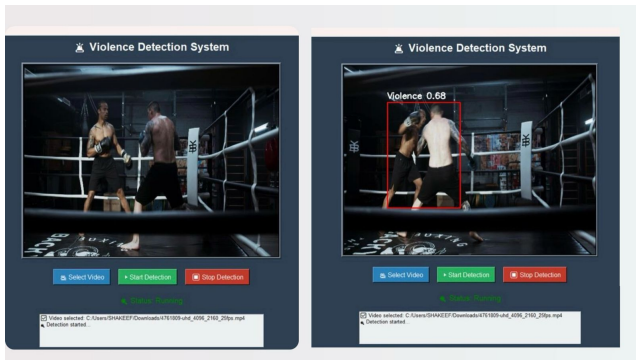


Figure 6: testing

Conclusion

- We have implemented a Human Violence Detection System using deep learning to enhance real-time surveillance and security.
- The system utilizes YOLOv8s for accurate detection of violent activities and weapon presence.
- Optimized deep learning architectures ensure high accuracy and real-time processing, enabling quick threat response.
- This project highlights the potential of AI-driven violence detection in improving public safety and reducing reliance on manual monitoring.

- The system can be enhanced by integrating gender detection to identify the gender of individuals involved in violent activities. This feature can help in generating gender-specific alerts and maintaining detailed records for better analysis and reporting.
- The system can be connected to IoT-enabled security devices like smart alarms, automatic doors, and surveillance drones, allowing automatic real-time responses during violence detection for faster prevention and action.
- The project can be extended to a cloud platform with a dedicated mobile application that enables remote monitoring, live notifications, and real-time alerting to users or authorities from any location.

References

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- [6] N. Jain, Van dana Gupta, U. Tariq, D. Jude Hemanth," Fast Violence Recognition in Video Surveil lance by In tegrating Ob ject Detection and

THANK YOU